

Lab 3: Architecture Selection & Justification

System: Community Lost & Found Matching Network

Architecture Selection: Layered Architecture was chosen for the Community Lost & Found Matching Network.

Reason 1: Clear Separation of Concerns Across a Cohesive Feature Set

The system's functionality : reporting, matching, claim verification, and moderation forms one cohesive application rather than independently scaling services. A layered structure (Presentation, Business, Data) lets the User Interface, Item Reporting, Matching Engine, Claim & Verification, and Moderation components evolve independently within the Business layer while sharing a single, consistent Database layer, without the operational overhead of managing distributed services.

Reason 2: Simplifies Data Consistency for Matching and Claims

Core use cases (UC-03: Match Items, UC-04: Claim Found Items) depend on strongly consistent reads of listing and claim status (for example, preventing a duplicate verified claim (BR-2) requires an authoritative, up-to-date view of claim records). A single, centralized Data layer avoids the eventual-consistency and cross-service coordination challenges that a microservices split would introduce for this workflow.

Reason 3: Security Advantage

The Claim & Verification component sits entirely within the Business layer and is the sole gatekeeper for revealing contact details (NFR-001). Because the Presentation layer has no direct access to the Data layer, the User Interface cannot bypass verification logic to read claim or contact data directly. Every request is forced through the Claim & Verification component's security-question check, reducing the attack surface for impersonation.

Reason 4: Performance Benefit

Isolating the Matching Engine within the Business layer, with a single required interface to the Database (ListingData), allows match computation to be optimized and, if needed, cached or indexed independently of the Presentation layer. This directly supports NFR-002: computing geo-proximity matches within 5 seconds at 10,000 active listings, without requiring changes to the User Interface or the Data schema